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INTERNATIONAL QUALIFICATIONS

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International AS and A-level

Chemistry 9620
CH01 Inorganic 1 and Physical 1
Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL AS CHEMISTRY (9620) CH01 – JANUARY 2025

The paper discriminated well.
Answers to numerical questions were often better than those requiring explanations.
Students were not always able to give definitions of key terms used in the specification and were not always familiar with practical techniques used in the experiments in the practical handbook. Students often did not give balanced equations and often failed to give state symbols when asked for them.

QUESTION 01

Most students gave the correct relative charges in part 1, but the relative masses were often incorrect. Part 2 was answered well by most students; common errors included ^{32}S and ^{34}Si. About half the students could give both electron configurations in part 3; errors included S given as [Ne] $3s^2$ $3p^4$ and V^{2+} given as [Ar] $4s^2$ $3d^1$, [Ar] $4s^2$ $4p^1$ or [Ar] $3d^5$. Answers to part 4 were generally good; errors included stating that proton number was different in the first part or stating that they had the same number of electrons in the outer shell in the second part.

QUESTION 02

Many students were unable to state the crystal structure of carbon monoxide; common errors included covalent or macromolecular. In part 2, just over half the students scored all the marks. Errors in the definition included using heat instead of enthalpy and omitting independent of the route. In the calculation, several students failed to use the moles of one or more substances given in the equation; most students gained the last mark for dividing their answer by 2. Many students answered part 3 well; errors included failure to appreciate that carbon dioxide has two C=O bonds or that ½ mole of O=O bonds were broken and some students gave the initial expression as ΣBonds formed – ΣBonds broken = $\Delta_r H$
21% of students scored the mark in part 4; insufficient answers included stating oxygen was an element or was in its standard state with no reference to oxygen being the only molecule with an O=O bond. In part 5 just under half the students answered the question completely; many students did not give the required detail. Errors included van der Waals' forces are 'low', which is not the same as 'weak', and many failed to state that these forces are between molecules.

QUESTION 03

Most students gave the correct equation in part 1; the most common error was missing state symbols. Part 2 was generally well done; common errors included magnetic field or electronic field. Most students gave the correct answer to part 3. Many students made good progress with part 4. Errors in calculating the mass of an ion were often seen, with the commonest error being the failure to convert grams to kilograms. A few students struggled with the rearrangement of the expression to calculate the speed and a few students made errors dealing with powers of ten. Students found part 5 challenging; about one fifth of students gained the mark.

QUESTION 04

In part 1, the first mark was scored by many students; errors in the second mark included not writing about atomic size (oxygen is smaller did not gain credit). Students found part 2 demanding; CH_4 was a common wrong answer. About 66% of students gave full answers, in part 3, stating that the molecules were symmetrical, and the dipoles cancelled out; errors included reference to polar bonds cancelling or that the molecule was tetrahedral without explaining that this meant it was symmetrical. In part 4, about a third of students scored all the marks; the most common error was a structure with no lone pairs and a tetrahedral shape.

QUESTION 05

In part 1, many students used the term concordant for the titres to be chosen to calculate a mean titre; errors included using the mean of all four titres or stating that the mean of titres 2 and 4 were used without explaining why. Many students gained full marks in part 2. Common errors included not subtracting the mass of the carbonate group from the M_r or omitting the division by 2 to get the A_r of one atom of **Q**. Students found part 3 challenging; a few students identified the mistakes in the method, but the explanations of the problems caused were not specific enough to gain the marks. Many students answered part 4 correctly.

QUESTION 06

Students found part 1 very challenging; there were many incorrect observations including white solid and bright white light, and equations that were balanced often lacked state symbols (asked for in the stem). In part 2, about half of the students gained both marks; errors included equations that were not ionic or giving the wrong colour of the precipitate. In part 3 many pupils gave a correct equation but found the formula of the species more difficult; HOCl or Cl^- were common errors for the substance showing the green colour. The majority of students correctly answered part 4. The equation in part 5 was generally done well; errors in this question included not balancing the equation and giving incorrect observations. $NaSO_4$ was a common incorrect formula in the equation. Part 6 differentiated well; common errors for the formula of the gas were SO_2 and HI. Equations were often not balanced, and the role of the iodide ion was often given as an oxidising agent.

QUESTION 07

In part 1, many students failed to gain marks because of missing out important words. The attraction between the particles was often described as a force, the ions were not described as positive metal ions and the electrons were not described as delocalised. Many students thought manganese was Mg. most students correctly answered part 2. Many students gave the correct equation in part 3; errors included electrons on the right hand side of the equation instead of the left, electrons missing and wrong number of H_2O on the right hand side. The half equation in part 4 was given by over three quarters of students. In part 5 the errors included failing to gather the terms together (particularly MnO_4^{2-}) on the left-hand side of the equation and failing to cancel out electrons. Some students found dealing with the hydrogens challenging; some added hydrogen gas on the left hand side or did not balance the hydrogens to give $4H^+$ on the left and $2H_2O$ on the right. Part 6 differentiated very well. Just over half the students gained full marks in the calculation and many were able to access some of the marks; common errors included using wrong powers of 10 in the volume conversion, rounding numbers too early in the calculation (ie before the last step as three significant figures was asked for in the question), failing to use the 2:1 stoichiometry from the equation and multiplying by 197.1 (K_2MnO_4) instead of 158 for $KMnO_4$

